Speaking of Design Patterns...

Congratulations on your first pattern!

You just applied your first design pattern—the STRATEGY Pattern. That's right, you used the Strategy Pattern to rework the SimUDuck app.
Thanks to this pattern, the simulator is ready for any changes those execs might cook up on their next business trip to Maui.

Now that we've made you take the long road to learn it, here's the formal definition of this pattern:

> **The Strategy Pattern** defines a family of algorithms, encapsulates each one, and makes them interchangeable. Strategy lets the algorithm vary independently from clients that use it.

Use THIS definition when you need to impress friends and influence key executives.

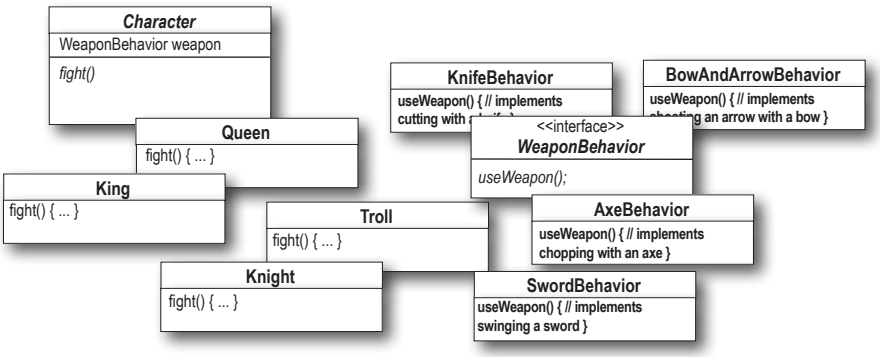
Design Puzzle

Below you'll find a mess of classes and interfaces for an action adventure game. You'll find classes for game characters along with classes for weapon behaviors the characters can use in the game. Each character can make use of one weapon at a time, but can change weapons at any time during the game. Your job is to sort it all out...

(Answers are at the end of the chapter.)

Your task:

1. Arrange the classes.
2. Identify one abstract class, one interface, and eight classes.
3. Draw arrows between classes.
   a. Draw this kind of arrow for inheritance ("extends").
   b. Draw this kind of arrow for interface ("implements").
   c. Draw this kind of arrow for HAS-A.
4. Put the method setWeapon() into the right class.




```
setWeapon(WeaponBehavior w) {
    this.weapon = w;
}
```

Overheard at the local diner...

Alice

I need a cream cheese with jelly on white bread, a chocolate soda with vanilla ice cream, a grilled cheese sandwich with bacon, a tuna fish salad on toast, a banana split with ice cream & sliced bananas, and a coffee with a cream and two sugars ... oh, and put a hamburger on the grill!

Flo

Give me a C.J. White, a black & white, a Jack Benny, a radio, a house boat, a coffee regular, and burn one!

What's the difference between these two orders? Not a thing! They're both the same order, except Alice is using twice the number of words and trying the patience of a grumpy short-order cook.

What's Flo got that Alice doesn't? **A shared vocabulary** with the short-order cook. Not only does that make it easier to communicate with the cook, but it gives the cook less to remember because he's got all the diner patterns in his head.

Design Patterns give you a shared vocabulary with other developers. Once you've got the vocabulary, you can more easily communicate with other developers and inspire those who don't know patterns to start learning them. It also elevates your thinking about architectures by letting you **think at the *pattern* level**, not the nitty-gritty *object* level.

Overheard in the next cubicle...

So I created this broadcast class. It keeps track of all the objects listening to it, and anytime a new piece of data comes along it sends a message to each listener. What's cool is that the listeners can join the broadcast at any time or they can even remove themselves. It is really dynamic and loosely coupled!

BRAIN POWER

Can you think of other shared vocabularies that are used beyond OO design and diner talk? (Hint: how about auto mechanics, carpenters, gourmet chefs, and air traffic controllers?) What qualities are communicated along with the lingo?

Can you think of aspects of OO design that get communicated along with pattern names? What qualities get communicated along with the name "Strategy Pattern"?

Rick, why didn't you just say you are using the **Observer Pattern**?

Exactly. If you communicate in patterns, then other developers know immediately and *precisely* the design you're describing. Just don't get Pattern Fever...you'll know you have it when you start using patterns for Hello World...

The power of a shared pattern vocabulary

When you communicate using patterns, you are doing more than just sharing LINGO.

Shared pattern vocabularies are POWERFUL. When you communicate with another developer or your team using patterns, you are communicating not just a pattern name but a whole set of qualities, characteristics, and constraints that the pattern represents.

"We're using the Strategy Pattern to implement the various behaviors of our ducks." This tells you the duck behavior has been encapsulated into its own set of classes that can be easily expanded and changed, even at runtime if needed.

Patterns allow you to say more with less. When you use a pattern in a description, other developers quickly know precisely the design you have in mind.

Talking at the pattern level allows you to stay "in the design" longer. Talking about software systems using patterns allows you to keep the discussion at the design level, without having to dive down to the nitty-gritty details of implementing objects and classes.

How many design meetings have you been in that quickly degrade into implementation details?

Shared vocabularies can turbo-charge your development team. A team well versed in design patterns can move more quickly with less room for misunderstanding.

As your team begins to share design ideas and experience in terms of patterns, you will build a community of pattern users.

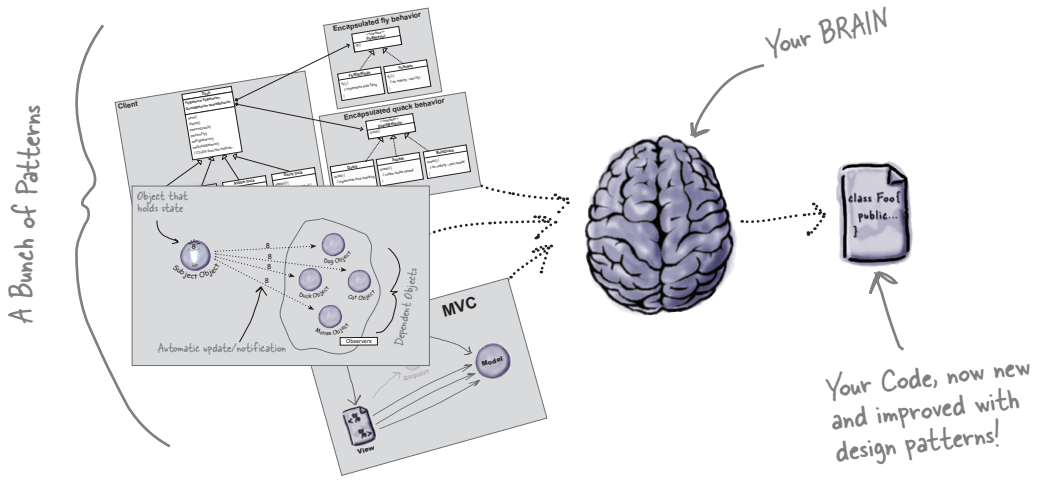
Shared vocabularies encourage more junior developers to get up to speed. Junior developers look up to experienced developers. When senior developers make use of design patterns, junior developers also become motivated to learn them. Build a community of pattern users at your organization.

Think about starting a patterns study group at your organization. Maybe you can even get paid while you're learning...

How do I use Design Patterns?

We've all used off-the-shelf libraries and frameworks. We take them, write some code against their APIs, compile them into our programs, and benefit from a lot of code someone else has written. Think about the Java APIs and all the functionality they give you: network, GUI, IO, etc. Libraries and frameworks go a long way toward a development model where we can just pick and choose components and plug them right in. But...they don't help us structure our own applications in ways that are easier to understand, more maintainable, and more flexible. That's where design patterns come in.

Design patterns don't go directly into your code, they first go into your BRAIN. Once you've loaded your brain with a good working knowledge of patterns, you can then start to apply them to your new designs, and rework your old code when you find it's degrading into an inflexible mess.



there are no Dumb Questions

Q: If design patterns are so great, why can't someone build a library of them so I don't have to?

A: Design patterns are higher level than libraries. Design patterns tell us how to structure classes and objects to solve certain problems, and it is our job to adapt those designs to fit our particular application.

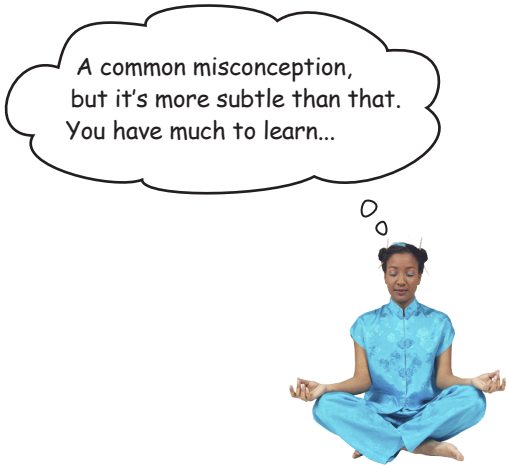
Q: Aren't libraries and frameworks also design patterns?

A: Frameworks and libraries are not design patterns; they provide specific implementations that we link into our code. Sometimes, however, libraries and frameworks make use of design patterns in their implementations. That's great, because once you understand design patterns, you'll more quickly understand APIs that are structured around design patterns.

Q: So, there are no libraries of design patterns?

A: No, but you will learn later about patterns catalogs with lists of patterns that you can apply to your applications.

Skeptical Developer



Friendly Patterns Guru

Developer: Okay, hmm, but isn't this all just good object-oriented design; I mean as long as I follow encapsulation and I know about abstraction, inheritance, and polymorphism, do I really need to think about Design Patterns? Isn't it pretty straightforward? Isn't this why I took all those OO courses? I think Design Patterns are useful for people who don't know good OO design.

Guru: Ah, this is one of the true misunderstandings of object-oriented development: that by knowing the OO basics we are automatically going to be good at building flexible, reusable, and maintainable systems.

Developer: No?

Guru: No. As it turns out, constructing OO systems that have these properties is not always obvious and has been discovered only through hard work.

Developer: I think I'm starting to get it. These, sometimes non-obvious, ways of constructing object-oriented systems have been collected...

Guru: ...yes, into a set of patterns called Design Patterns.

Developer: So, by knowing patterns, I can skip the hard work and jump straight to designs that always work?

Guru: Yes, to an extent, but remember, design is an art. There will always be tradeoffs. But, if you follow well-thought-out and time-tested design patterns, you'll be way ahead.

Developer: What do I do if I can't find a pattern?

Remember, knowing concepts like abstraction, inheritance, and polymorphism does not make you a good object-oriented designer. A design guru thinks about how to create flexible designs that are maintainable and can cope with change.

Guru: There are some object-oriented principles that underlie the patterns, and knowing these will help you to cope when you can't find a pattern that matches your problem.

Developer: Principles? You mean beyond abstraction, encapsulation, and...

Guru: Yes, one of the secrets to creating maintainable OO systems is thinking about how they might change in the future, and these principles address those issues.

Tools for your Design Toolbox

You've nearly made it through the first chapter! You've already put a few tools in your OO toolbox; let's make a list of them before we move on to Chapter 2.

OO Basics

Abstraction

Encapsulation

Polymorphism

Inheritance

We assume you know the OO basics like abstraction, encapsulation, polymorphism, and inheritance. If you are a little rusty on these, pull out your favorite object-oriented book and review, then skim this chapter again.

OO Principles

Encapsulate what varies.

Favor composition over inheritance.

Program to interfaces, not implementations.

We'll be taking a closer look at these down the road and also adding a few more to the list.

OO Patterns

Strategy – defines a family of algorithms, encapsulates each one, and makes them interchangeable. Strategy lets the algorithm vary independently from clients that use it.

Throughout the book, think about how patterns rely on OO basics and principles.

One down, many to go!

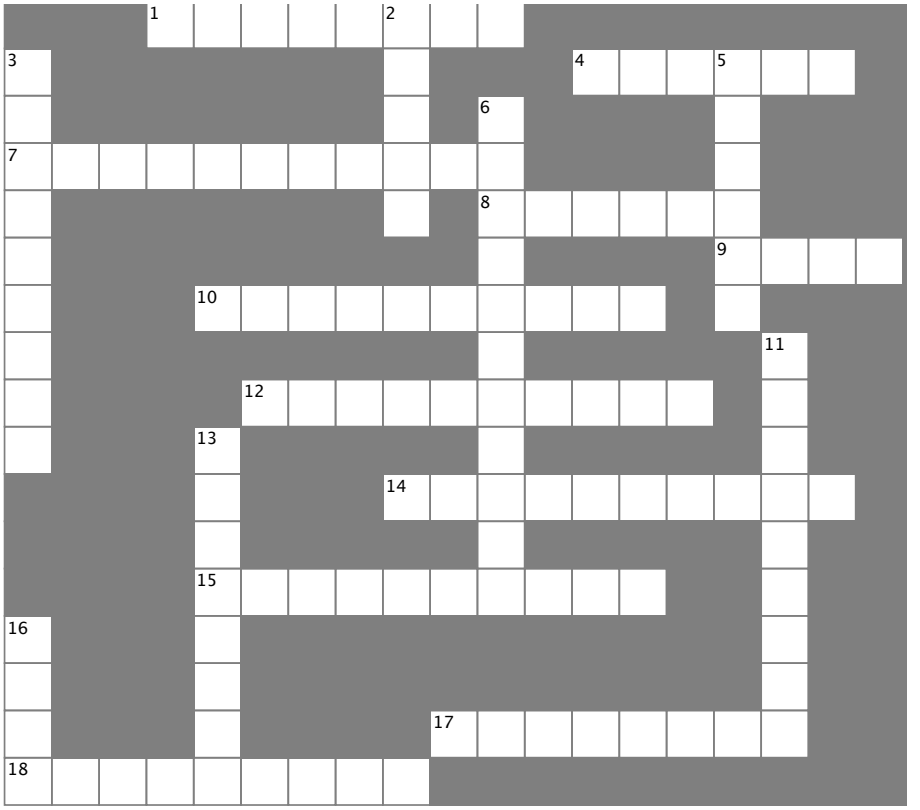
BULLET POINTS

- Knowing the OO basics does not make you a good OO designer.
- Good OO designs are reusable, extensible, and maintainable.
- Patterns show you how to build systems with good OO design qualities.
- Patterns are proven object-oriented experience.
- Patterns don't give you code, they give you general solutions to design problems. You apply them to your specific application.
- Patterns aren't *invented*, they are *discovered*.
- Most patterns and principles address issues of *change* in software.
- Most patterns allow some part of a system to vary independently of all other parts.
- We often try to take what varies in a system and encapsulate it.
- Patterns provide a shared language that can maximize the value of your communication with other developers.

Design Patterns Crossword

Let's give your right brain something to do.

It's your standard crossword; all of the solution words are from this chapter.



ACROSS

1. Paatterns can help us build _____ applications.
4. Strategies can be _____.
7. Favor this over inheritance.
8. Development constant.
9. Java IO, Networking, Sound.
10. Most patterns follow from OO _____.
12. Design patterns are a shared _____.
14. High-level libraries.
15. Learn from the other guy's _____.
17. Pattern that fixed the simulator.
18. Program to this, not an implementation.

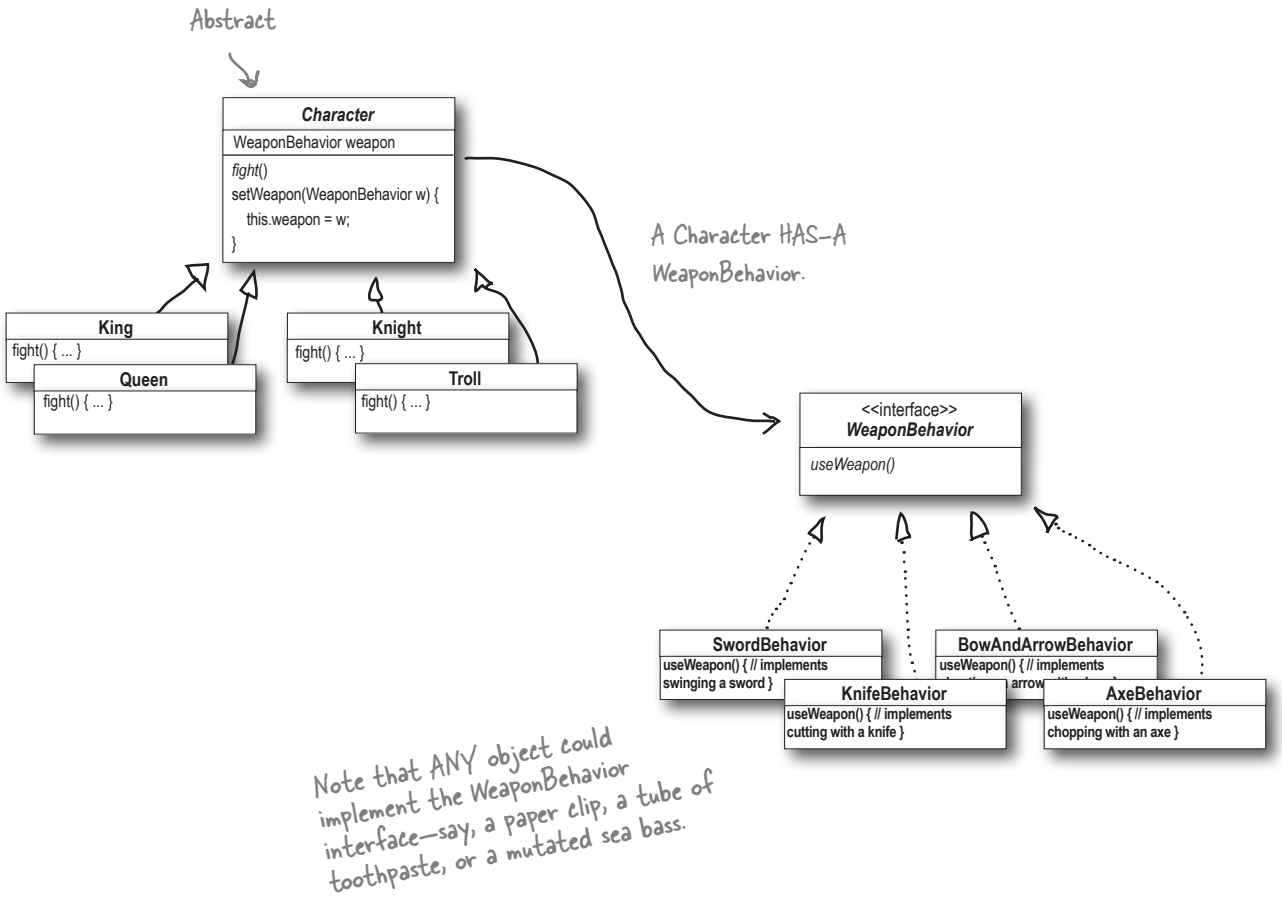
DOWN

2. Patterns go into your _____.
3. Duck that can't quack.
5. Rubber ducks make a _____.
6. _____ what varies.
11. Grilled cheese with bacon.
13. Rick was thrilled with this pattern.
16. Duck demo was located here.

Design Puzzle Solution

Character is the abstract class for all the other characters (King, Queen, Knight, and Troll), while WeaponBehavior is an interface that all weapon behaviors implement. So all actual characters and weapons are concrete classes.

To switch weapons, each character calls the setWeapon() method, which is defined in the Character superclass. During a fight the useWeapon() method is called on the current weapon set for a given character to inflict great bodily damage on another character.



Sharpen your pencil Solution

Which of the following are disadvantages of using subclassing to provide specific Duck behavior? (Choose all that apply.) Here's our solution.

- ☑ A. Code is duplicated across subclasses.
- ☑ B. Runtime behavior changes are difficult.
- ☐ C. We can't make ducks dance.
- ☑ D. It's hard to gain knowledge of all duck behaviors.
- ☐ E. Ducks can't fly and quack at the same time.
- ☑ F. Changes can unintentionally affect other ducks.

Sharpen your pencil Solution

What are some factors that drive change in your applications? You might have a very different list, but here's a few of ours. Look familiar? Here's our solution.

My customers or users decide they want something else, or they want new functionality.

My company decided it is going with another database vendor and it is also purchasing its data from another supplier that uses a different data format. Argh!

Well, technology changes and we've got to update our code to make use of protocols.

We've learned enough building our system that we'd like to go back and do things a little better.

Design Patterns Crossword Solution

			1 F	L	E	X	I	2 B	L	E							
3 D								R				4 R	E	U	5 S	E	D
E								A		6 E					Q		
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